Week 2: Quality Attributes & Scenarios

Analysis and Assessment of Design Quality

Today's learning goals

1. Use quality attributes as evaluation lenses, not slogans
2. Write structured quality scenarios with response measures
3. Surface attribute conflicts and non-goals deliberately
4. Link attributes to observable design decisions and failure modes

Agenda ($\approx$ 2 hours)

1. Attribute families (practical subset)
2. From slogan to lens
3. Scenario anatomy (Bass-style, usable)
4. Conflicts and non-goals
5. Mapping attributes $\leftrightarrow$ design evidence
6. Practice preview

A practical attribute set

We lean on ISO/IEC 25010–style families without memorising the whole standard:

- Performance efficiency — latency, throughput, resource use
- Reliability — availability, fault tolerance, recoverability
- Security — confidentiality, integrity, authenticity, accountability
- Maintainability — modularity, analysability, modifiability, testability
- Usability — learnability, operability, accessibility (as relevant)
- Compatibility / interoperability and portability when the brief needs them

Scalability often appears as a scenario family across performance and reliability.

Attributes are lenses, not badges

Weak: “The system is secure and scalable.”

Stronger: “Under exam-week login storm (stimulus), p95 login < 800 ms with auth service degraded gracefully (response + measure).”

Remember

If you cannot invent a failing test or observable failure, the attribute claim is not yet evaluable.

Quiz 1 — Attribute identification

Quiz

Scenario: “When a paid invoice PDF fails to generate, the user still sees order confirmation within 2 s, and ops is alerted within 1 min.” Which attribute pair is primarily under test?

1. Usability and portability
2. Reliability (graceful degradation / recoverability) and a slice of performance
3. Only maintainability
4. Only security confidentiality

Pause — discuss with a neighbour or answer privately before the next slide.

Answer — Quiz 1

Answer

2. Failure handling + user-visible continuity = reliability; the 2 s bound adds a performance constraint on the happy/degraded path.

Scenario anatomy (practical)

Adapted from classic quality-attribute scenario parts:

1. Source — who/what triggers (user, timer, attacker, peer service)
2. Stimulus — the event
3. Environment — mode (normal, peak, degraded, recovery)
4. Artifact — what is stimulated (system, channel, data store)
5. Response — what should happen
6. Response measure — measurable / observable criterion

MSc note

You do not need full ATAM trees; you do need response measures.

Worked example — maintainability

Source	Developer
Stimulus	Change VAT rules for one country
Environment	Normal development; regression suite green required
Artifact	Billing module + pricing rules
Response	Change isolated; no UI rewrite
Measure	$\leq$ 1 module + 1 config touched; tests updated same day

Worked example — security

Source	External client with stolen user token
Stimulus	Attempt to export another user's grades
Environment	Production API
Artifact	Grades export endpoint
Response	Deny; audit event written
Measure	0 unauthorised rows; audit within 5 s; alert if N failures / min

Vague vs testable

Vague	Testable direction
“Make it fast”	p95 search < 300 ms at 200 RPS
“Be secure”	No PII in logs; admin actions audited
“Easy to change”	Feature X change touches ≤ 2 packages
“Handle failures”	Read-only mode if payments down

Quiz 2 — Missing scenario part

Quiz

"When traffic spikes, the system should remain available." What is the most critical missing piece for evaluation?

1. A fashionable CDN brand name
2. Response measure (and clearer stimulus/environment): how much spike, what availability
3. A UML sequence diagram
4. Proof that microservices are used

Pause — discuss with a neighbour or answer privately before the next slide.

Answer — Quiz 2

Answer

2. Without magnitude, environment, and measure, reviewers cannot agree on pass/fail.

Attributes conflict — treat that as signal

Classic tensions:

- Security checks vs latency / usability
- Consistency vs availability / partition tolerance (distributed)
- Modularity vs short-term delivery speed
- Observability verbosity vs privacy / cost

A review that pretends there is no conflict is usually incomplete.

Non-goals are first-class

Explicit non-goals prevent false findings:

- “Not targeting multi-region active-active in v1”
- “Admin UI accessibility deferred to v1.1”
- “Batch reports may take minutes; interactive path must not”

Common pitfall

Filing a “smell” against a documented non-goal without arguing why the non-goal is unsafe.

Quiz 3 — Conflict handling

Quiz

Brief prioritises auditability of every admin action and sub-second UX for admins. A design batches audits asynchronously (eventual). Best evaluation stance?

1. Automatic fail—async audit can never be correct
2. Accept silently—performance always wins
3. Accept only if loss/delay bounds and alerting are specified relative to the auditability scenario; else major finding
4. Demand both sync audit and sub-ms latency with no trade-off discussion

Pause — discuss with a neighbour or answer privately before the next slide.

Answer — Quiz 3

Answer

3. Tie the mechanism to the scenario's response measure; require bounds. Dogmatic fail/accept ((1)/(2)) skips analysis; (4) denies physics and process.

From attribute to design evidence

Attribute focus	Example design evidence
Maintainability	Module boundaries, ownership, test seams
Performance	Caching, sync vs async paths, data locality
Reliability	Timeouts, retries with jitter, bulkheads, degraded modes
Security	Trust boundaries, authn/z points, data classification

Evaluation

asks: does the structure make the scenario plausible?

Utility tree — light use only

Optional organising tool (ATAM-inspired, shallow):

1. List attributes
2. Attach 1–3 scenarios each
3. Rank by business importance × technical risk (H/M/L)

We do not run full ATAM workshops in this course—but ranking scenarios guides review time.

Quiz 4 — Attribute ↔ structure

Quiz

Maintainability scenario: weekly rule changes must stay in one module. The design puts rules in UI widgets, a shared “utils” jar, and stored procedures. Best finding?

1. Style nit: rename utils
2. Major maintainability risk: shotgun change across layers; recommend a single rules owner
3. Performance defect by definition
4. Non-issue because three locations show “flexibility”

Pause — discuss with a neighbour or answer privately before the next slide.

Answer — Quiz 4

Answer

2. Evidence of shotgun change directly violates the scenario's measure.

Key takeaways

- Attributes without scenarios are slogans
- Response measures make reviews decidable
- Conflicts and non-goals belong in the evaluation record
- Structure is evidence for (or against) scenarios

Practice today

Handout: 02-week02-practice.md
Top-3 attributes, three full scenarios, conflict matrix.

Next week

Design smells & structural health
Architectural significance, coupling/cohesion, mitigations vs rewrite fantasies.